

Alexander Elliott

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TECHNOLOGIES SolidWorks, CATIA, Fusion 360, Onshape, Revit, AutoCAD, Arduino, Raspberry Pi, Docker, Soldering, 3D Printing, Mach 4, EtherCAT
LANGUAGES Java, Python (NumPy, Pandas, OpenCV, TensorFlow), C, C++, C#, MATLAB, SQL, JavaScript (React, Three.js), HTML/CSS

University of British Columbia, Canada, BAsc Mechatronics (COOP)

SEP 2021 - MAY 2026

Cumulative GPA: **3.8**, Dean's List: **4 Years**

Relevant Coursework: **Thermal & Fluid Dynamics, Circuits, Microcomputers, Firmware, Algorithms, Operating Systems**

RELEVANT EXPERIENCE

Magna International — Mechatronics CO-OP (CAE Team)

JAN - AUGUST 2025

- Developed automation tools for pre- and post-processing of simulations (**FEA**, **MBD**, **Inertial Analysis**) using **TypeScript** and **Python** tech stacks, streamlining the workflow for the global CAE teams
- Architected and deployed "Inertia+", a tool that automated component **inertia** calculations and report generation, leading to upwards of **26%** time savings (**270+** hours saved per quarter)
- Spearheaded the creation of "Collab Space," a web-based platform using **JavaScript** and **Python** to visualize **FEA** results, enabling real-time collaboration and faster design iteration

AME Group — Mechanical CO-OP, QAQC CO-OP

SEPTEMBER - DECEMBER 2023, MAY - AUGUST 2024

- Designed 15+ HVAC, Hydronic, and Plumbing layouts and schematics for mixed residential spaces in the Vancouver area using **Revit** and AutoCAD
- Optimized AutoCAD schedule/schematic creation using **Python** and **Revit** plugins to reduce repetitive work by 50%
- Wrote C# Macros to automate **Revit** tasks such as creating detailed schematics from MEP systems, simplifying drafting tasks, and converting **AutoCAD** details
- Led many BIM improvement projects for teams across AME, such as **Revit** Schematics and **Revit** Details, to reduce the need for **AutoCAD**, allowing the user to work solely in **Revit**, saving dozens of hours on 5+ projects

UBC Uncrewed Aircraft Systems — Mechanical Payload Team

SEPTEMBER 2022 - SEPTEMBER 2024

- Prototyped various payload drop systems, exploring technologies such as electromagnetic clutches and guided parachutes to control water bottle drops safely using **Onshape**
- Designed and manufactured a Carbon Fibre enclosure for our scale model taxi enclosure
- Achieved 2nd place in Phase 1 and 3rd place in Phase 2 of the 2024 AEAC competition

LRA3D — Mechanical Designer / Fabricator Intern

MAY - AUGUST 2023

- Custom-designed prop/set installations for 3 clients using Rhino 7, including a 20' tall plywood guitar prop
- Scripted presets for **CNC** Routers and Hotwire machines using Mach 4 to increase machine efficiency by 20%
- Wired **CNC** Hotwire machine with **EtherCAT** technology to modernize production, reducing the CPU load by 30%

PROJECTS

Spice Changer — UBC Mech 423 Project

JANUARY - FEBRUARY 2026

- Created an **automated** kitchen spice rack to aid chefs in the kitchen, using a 3D printed rotating disk design to speed up spice retrieval and storage
- Designed a **Windows Form C#** UI to implement fast and reliable controls, including voice control and restocking features
- Implemented controls on an **MSP430** chip on a custom PCB to drive the **stepper motor** to push and pull the spices, and a **DC motor** to align the rotating spice tray to the correctly requested spice

LapSim Optimizer Tool — UBC Formula Electric Mechanical Team

MAY - AUGUST 2024

- Engineered a lap time simulation tool in **MATLAB** using vehicle dynamics models to predict competition performance within **75%**
- Influenced key design decisions for the vehicle's powertrain and aerodynamics by providing data-driven estimates of how parameter changes would impact competition scores

Interests and Hobbies — Video Editing, Robotics, Guitar, Photography, Skiing, Biking, Geography, Space, Sneakers